

The Versatile Rigging of Splat in "Strange World"

Cameron Black
Walt Disney Animation Studios
Burbank, USA
Cameron.Black@disneyanimation.com

Christoffer Pedersen
Walt Disney Animation Studios
Burbank, USA
Chris.Pedersen@disneyanimation.com

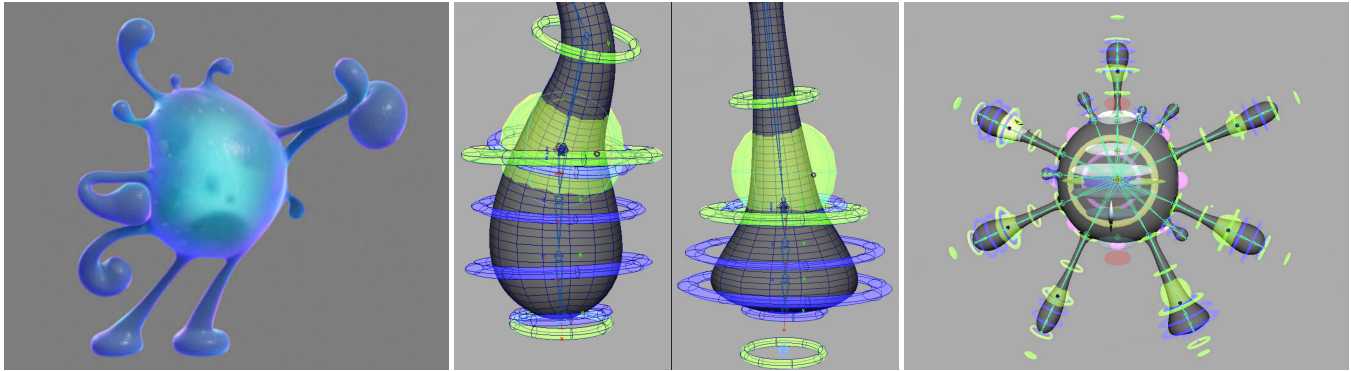


Figure 1: Final rendered pose and character rig examples. ©2023 Disney

ABSTRACT

The physical and kinetic complexities presented by the character Splat in Walt Disney Animation Studios' recent film "Strange World" are unlike anything the studio typically encounters from a rigging perspective. Splat's unique physiology - an omni-directional character setup where up could become down and arms and legs were completely interchangeable - presented many unique obstacles for the character teams. In order to meet the challenges presented by the scope of Splat, new rigging systems were created, enabling limbs to be used for a large variety of situations, as well as an overall versatility of the entire rig to help support unanticipated uses in the film.

CCS CONCEPTS

• Computing methodologies → Animation.

KEYWORDS

character, rigging, animation

ACM Reference Format:

Cameron Black and Christoffer Pedersen. 2023. The Versatile Rigging of Splat in "Strange World". In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595450>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Talks, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0143-6/23/08.

<https://doi.org/10.1145/3587421.3595450>

1 INTRODUCTION

Despite a lack of face, Splat is one of the most expressive characters created in "Strange World." From the start of production, it was apparent that this character would be both important to the story telling and very complex. The story and art department set the bar high during development, which required the character team to develop new and unique ways to deliver on this potential. Ultimately, Splat's performance would rely heavily on pantomime and extremely exaggerated movements. This was an equally difficult task for animation as it was for rigging.

2 VERSATILE RIGGING

While many of the required actions and poses Splat would need to hit during the film were known, there were also seemingly endless possible unknown requirements which would naturally come as animation applied their creativity to the shots, and sparked new ideas about Splat's performance. The first issue that needed to be addressed on Splat was the omni directionality of its performances. From the beginning, it was well known that Splat would need to be pliable, flip around and would also never have a true "up" or "down" side. This can be demonstrated through several methods

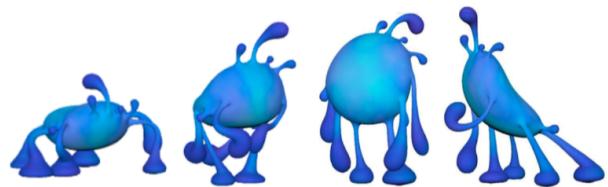


Figure 2: Splat in multiple locomotive orientations. ©2023 Disney

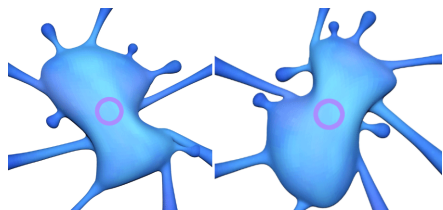


Figure 3: Splat shape controls oriented around the body.
©2023 Disney

of locomotion in Figure 2. In the film, Splat can be seen spinning horizontally while swapping which limb is in contact with the ground, as well as walking and running as a quadruped would.

2.1 Omni-Directional Orientation

After many brainstorming sessions, it was decided that the best approach would be to split Splat into multiple pieces and rig them individually, then have the final mesh be driven by all the components. Using this approach, the body was rigged separately. An FK chain was created that would give Splat the ability to fold, similar to the way you might rig a clamshell. Where this setup sets itself apart from the standard IK/FK rig is in the binding. The mesh is not driven by a skincluster, but rather a rig lattice. The advantage of this is the ability to layer additional deformers onto this lattice cage, enabling the lattice and FK rig to be oriented on the fly without deforming the original geo. This empowers the animator to choose the orientation and angle of which to bend the FK spine. While very simple, this setup allowed for a huge range of motion to start with. Another instance of a rotatable setup is the silhouette shapers. This was responsible for overall shaping of the body, allowing it to warp and squeeze as shown in Figure 3, or be used to fine tune its silhouette. Instead of using a lattice like the previously mentioned system, this setup used a proprietary wire deformer with a live bind feature.

2.2 Multi use limbs

The next problem that needed to be addressed was how to approach the limbs and the ability to use each as an arm, leg, or a tentacle. This would also require a method for blending between both a bendy curvy shape and one with hard corners for various situations. For situations where the limbs were legs, simulating the feel of compression on the squishy, bulbous tips as they contact the ground was required. To achieve this, the existing binding of the simplified joint structure of the tips was leveraged, allowing for animation to control this feature either with the free-form tip IK control, or with a simplified attribute which was most useful for walking situations. This feature was also beneficial for when the limbs were "arms" and were required to push up against objects, as seen in Figure 1. For situations where the limbs were hands, they were required to do the usual hand things like, point, grasp, and emote. Movable pivots were added so the tips could roll on different objects as needed, or pivot from any position to help achieve the feel of a hand with this simplified shape. Additionally, an intuitive roll setup was incorporated, a rig system usually reserved for characters with feet to easily facilitate ball to toe foot rolls.

2.3 "Removable" limbs

There were situations where some limbs needed to be removed completely in shots to simplify a pose. This ability was given to the smaller limbs of Splat. The topology was locked, so actually removing the limbs on the fly was not an option. When the limb was turned off, the topology was smooshed down to contour the body. While simple in concept, the challenge was to not see any sort of artifacting of the limb that collapsed to the body, especially when that area was deformed. This shape change was carefully layered into the deformation stack to work best with the other deformers. Once implemented, animation was able to toggle these limbs off as needed, seemingly removing it completely from the rig.

2.4 Stitching it all together

Once the rigging of all the individual parts was complete, they had to be stitched together. This was made easy using the rig component system. The rigs were able to be rebuilt and combined using the internal components system, build around dRig[Smith et al. 2012]. With this proprietary tool, blocks of code were easily instanced and reused from parent components in a quick easy manner. Leveraging this, each limb was glued to the body using our bindmesh component. A bindmesh is a small, single faced plane that is typically bound to the same skincluster as the body geometry in a rig. But in this case, the mesh is driven by the binding lattice driving the body geo. One of these bindmeshes were placed at the root position of each tentacle. A rivet was connected to the bindmesh center, serving as the main driver of the tentacle. Once the tentacles were attached to the main body geo, all the pieces were bound together, which, in turn, drove the final render mesh.

3 CONCLUSION

The performance requirements of Splat in "Strange World" presented the rigging team with numerous complex problems. Versatility was front of mind as rig systems were designed to free animators of a rigidly established rig orientation. This, combined with a multi-use limb and the ability to remove smaller limbs on the fly gave splat the flexibility to keep up with the imagination of its animators. Creating modular rig systems that allowed for easier implementation and modification of rig setups was critical to the success in designing this unique and complex character rig.

REFERENCES

- Greg Smith, Mark McLaughlin, Andy Lin, Evan Goldberg, and Frank Hanner. 2012. dRig: An Artist-Friendly, Object-Oriented Approach to Rig Building. In *ACM SIGGRAPH 2012 Talks* (Los Angeles, California) (SIGGRAPH '12). Article 18, 1 pages. <https://doi.org/10.1145/2343045.2343069>